Agrivoltaics
AI Drone Senior Project

Abstract

Image/video processing AI to determine when and which crops need sprayed with pesticide. Utilizes a camera which can be mounted on existing farm equipment and sends data to the cloud. Data then sent by AI model and results displayed in application. Additional stationary sensor boxes provide zone specific alerts and frost prediction. Stationary sensor boxes utilize solar energy to maintain battery charge for data collection.

Purpose

Provide a reliable network of agricultural monitoring devices that enables low-subsistence farmers to make data-driven decisions, improve crop health, and increase yield. Real-time environmental data supports efficient resource use, including applying pesticides only when necessary, reducing chemical use and associated costs while protecting plant health.

Technologies Utilized

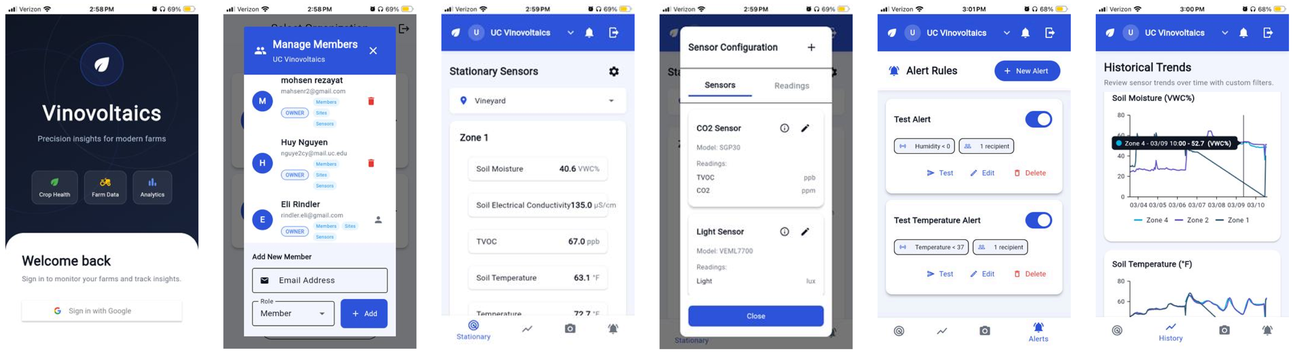
Firebase
PyTorch
Google BigQuery
Flutter
docker

Sponsors

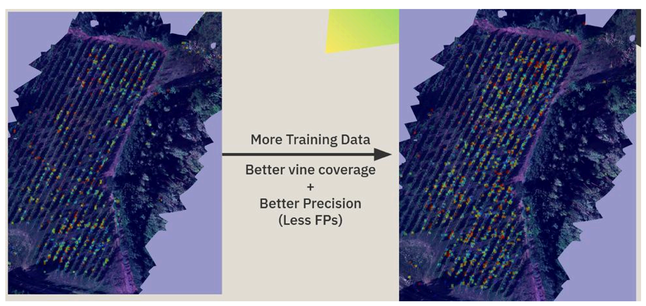
UC
omid

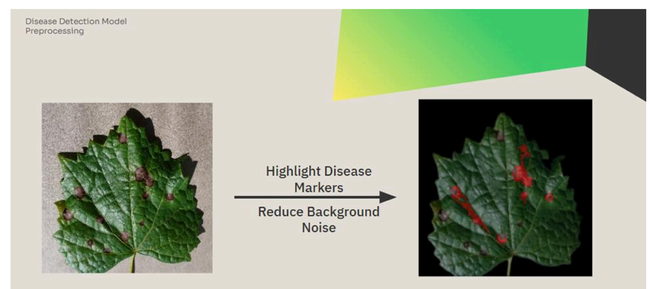
Notable changes / Accomplishments

Improved User Interface, requiring authentication for users and allowing customization of the deployed configuration (alerts, prediction windows, sensor types and zone counts)



The original Disease model reported a roughly 80% accuracy on lab data. New and improved version achieves > 96% accuracy on the same lab data. This was achieved through training larger capacity models (Resnet 50, ResNet 101, ResNet152, efficientNetB4) which in turn distill their knowledge to a smaller capacity model, the original ResNet18. This was to ensure we got the most out of out limited training data, while attempting to combat overfitting.





A Vision Tensor (ViT) model was fine-tuned to classify commonly occurring diseases using close up images of grape leaves with a 97% accuracy on test data. By tokenizing images into 16 x 16 patches and using Multi-Head Self-Attention, the model identifies subtle disease markers that traditional convolutions might miss. An additional model was devised to allow the farmer to use their smartphone camera to detect the type of pests present on vines.

Likewise, a frost prediction model is deployed and running in the cloud, attempting to guess the chance of frost in the proceeding 6 hour window, for each zone sensor box. This prediction is restricted by the farmer to a relevant inference date window, and only when the temperature falls below the configured threshold (example: September through November when < 39° F)

Future Improvements

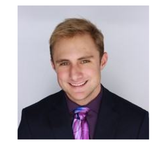
This is an ongoing project that will be passed on to next year's team. Major improvements needed include (but are not limited to):

- Moving training pipelines to a dedicated, cost managed, cloud environment for efficiency
- Collection of more training data during growing season from the multi-spectral camera
- Inclusion of reinforcement learning for farmer feedback on model outputs
- Add more sensors to the stationary sensor box to increase data capabilities
- Incorporate a PCB to create stable electrical connections within sensor boxes

Meet the Team

Eli Rindler

Degree: Computer Science
Subteam: Web Application



Nick Tatko

Degree: Mechanical Engineering
Subteam: Drone & Sensor Boxes

Dylan Serger

Degree: Computer Science
Subteam: AI Models

Huy Nguyen

Degree: Computer Science
Subteam: Mobile Application

Ayushi Sharma

Degree: Computer Science
Subteam: AI Models

Amogh Silaparasetti

Degree: Computer Science
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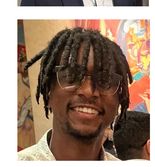
Kyle Sweeney

Degree: Electrical Engineering
Subteam: Drone & Sensor Boxes

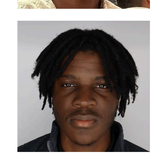
Cliff Sheehan

Degree: Mechanical Engineering
Subteam: Drone



Julius Dickerson

Degree: Electrical Engineering
Subteam: Sensor Boxes



Jesubamigbe Bamgboye

Degree: Electrical Engineering
Subteam: Sensor Boxes

Project Advisor

Dr. Mohsen Rezayat

Founder and member of the Board of Directors (BOD) of poverty-alleviation non-profit OMID USA organization